EPM213s - Electrical Measurements

AC Measurement Using PMMC (With Rectifiers)

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AC Measurement Using PMMC (With Rectifiers)

1. Objective

To measure AC voltage or current using a PMMC instrument with a rectifier circuit.

2. Apparatus Used

- - PMMC Instrument
- - Diodes (Silicon)
- - Resistors
- - AC Power Source
- - Digital Multimeter
- - Breadboard and connecting wires

3. Theory

PMMC instruments can only measure DC quantities. To measure AC signals, the signal must be rectified into DC using diode rectifiers.

A full-wave rectifier is typically used to convert the AC into pulsating DC, which can then be measured by the PMMC.

Key Concepts:

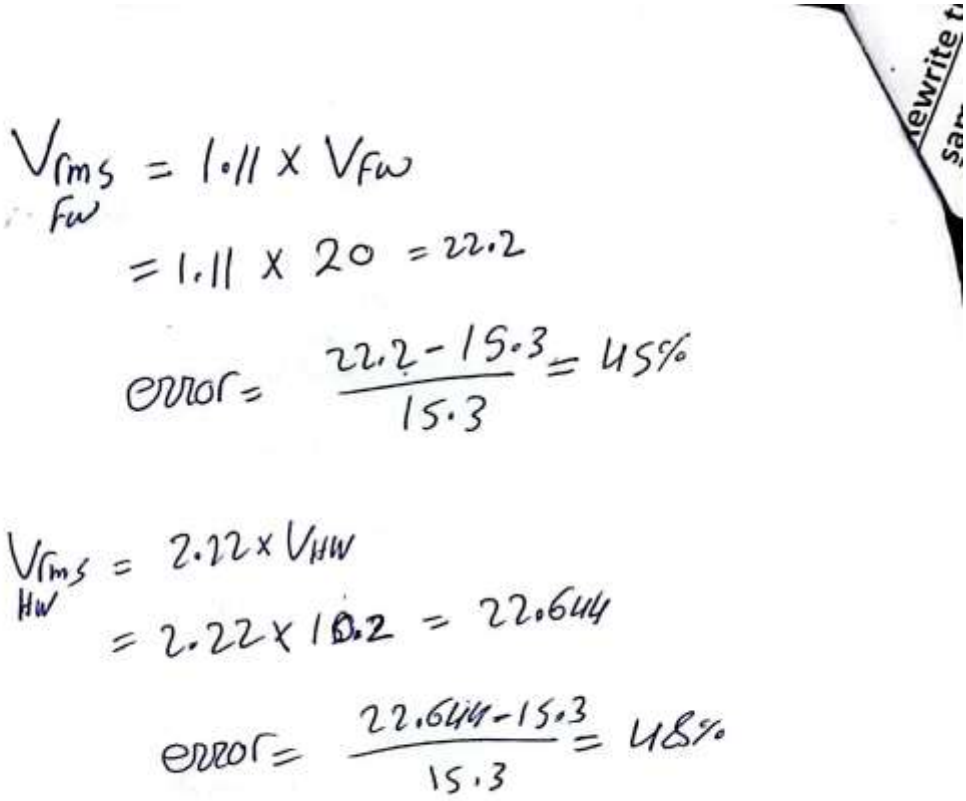
- **Diode Rectification**: Allows current to flow in one direction, converting AC to DC.

- **Average or RMS Calibration**: The meter is often calibrated in RMS values even though it measures average DC after rectification.

4. Procedure

1. Construct a full-wave rectifier circuit using diodes.
2. Connect the PMMC instrument at the output of the rectifier.
3. Apply a known AC voltage source to the input.
4. Observe the deflection of the PMMC pointer.
5. Compare the reading with a true RMS value measured by a digital multimeter.
6. Repeat the experiment with varying AC voltages.

5. Observations and Calculations



7. Conclusion

By using a rectifier circuit, the PMMC instrument was able to measure AC voltages. Though the instrument reads average DC voltage, with proper calibration it can effectively indicate RMS values of AC signals. This method enables PMMC usage in AC measurements.

8. Photos of Experimental Setup

